

SwiftBite Delivery — Zone-Hour Incentive: Experiment Read-Out

The exhaustive record behind the board summary · dual-track (pandas + DuckDB)

The question and the answer

Question. SwiftBite ran a randomized zone-hour incentive (a peak-block per-delivery bonus, unit = zone-day) to fix supply-short liquidity. Does it work, is it safe, and should it roll out?

Answer. It **improves liquidity** — fulfillment +1.87 pp, ETA p90 -6 min, no-courier cancels -0.36 pp, robust across cluster-robust, randomization-inference and wild-cluster-bootstrap inference, concentrated in the supply-short and balanced zones (interaction $p = 0.004$), with only a small (~17%, not significant) neighbour-zone drag. It **does not pay for itself**: ~R\$ 281 per incremental delivered order vs a ~R\$ 10 contribution margin, because the flat bonus subsidises every delivery while only ~2% are incremental. **Recommendation: don't ship the flat bonus; restructure it to reward incremental supply and cap it to the short + balanced zones.**

Appendix — the eight analyses in full

01 — Anchor: the experiment frame and the naive numbers

Question. What is the analysis frame, and what do the raw treated-vs-control differences look like before any modelling?

Method (both tracks). Load `fact_experiment_zone_day` (one row per randomised zone-day, peak block) joined to `fact_incentive_assignment` (design + frozen pre-period covariates) and `dim_zone` (baseline supply-stress tier). Group by arm; group by tier × arm. DuckDB reproduces the headline differences on the same parquet.

Evidence.

	Control (n=420)	Treatment (n=420)
Fulfillment rate	95.75%	97.61%
ETA p50 / p90 (min)	36.8 / 46.2	33.9 / 40.1
No-courier cancel rate	1.16%	0.80%
Courier earnings / active hr	R\$ 33.0	R\$ 41.2
Incentive spend	R\$ 0	R\$ 35,820

Naive fulfillment lift **+1.87 pp**; ETA p90 **−6.06 min**; no-courier **−0.36 pp**.

Naive fulfillment lift by tier: short **+2.29 pp**, balanced **+2.43 pp**, long **+0.94 pp**.

Conclusion. 840 zone-days, exactly 420/arm, spread over 12 zones (6 short / 2 balanced / 4 long) and 70 dates. Every treated-vs-control gap points the same way — more orders delivered, faster, fewer no-courier cancels, more courier earnings — and the gap is visibly larger in the supply-short and balanced zones than in the long (well-supplied) zones. The rest of the read-out establishes whether those gaps are causal (03), safe on the guardrails (04), stable over time (05), genuinely heterogeneous (06), and not just supply pulled from neighbouring zones (07).

Limitations. The naive by-arm means ignore the clustering of zone-days within zones — they are a description, not an estimate. Standard errors and inference start in 03 .

02 — Randomisation & balance: can the assignment be trusted?

Question. Was the zone-day assignment balanced, and is there any hidden selection to worry about?

Method (both tracks). (1) χ^2 sample-ratio test on treated/control counts — overall, per stress tier, per weekday/weekend. (2) Standardized mean differences (treatment – control) on the frozen pre-period (months 1–15) covariates: `pre_fulfillment_rate`, `pre_eta_p90_min`, `pre_orders_placed_mean`, `pre_liquidity_ratio`, `pre_idle_courier_ratio`, plus `is_weekend`. A Love plot.

Evidence (`outputs/tables/02_srm.csv`, `02_balance.csv`, `outputs/figures/02_love_plot.png`).

Cut	χ^2	p
overall · per tier · per weekday/weekend	0.0	1.00

All standardized mean differences are **0.000**.

Conclusion. The allocation is **exactly 50/50 within every (zone × weekday/weekend) cell** — it was drawn that way (stratified, balanced on day-type), so the χ^2 is degenerate and passes trivially. The covariate SMDs are exactly zero because the pre-period covariates are **zone-level** (one frozen value per zone) and each zone is split 50/50, so the two arms carry identical covariate means by construction.

This means the two classic A/B risks do **not** apply here: there is no sample-ratio mismatch to investigate, and there is no differential attrition — a zone-day cannot "drop out". What *could* still bias the estimate is a **time-varying shock** that lands differently on treated vs control zone-days (weather, a local event). Those enter the analysis in `03` as covariates (`is_rainy`, `has_local_event` on `fact_zone_hour`) and via the concurrent control arm; the day-of-week balance above rules out the largest such confounder.

Limitations. Balance on *observed* zone-level covariates only; the design's real protection is the within-zone split and the concurrent control, not covariate adjustment.

03 — Primary effect: does the zone-hour incentive lift fulfillment?

Question. Causal effect of arming the peak-block bonus on the **fulfillment rate** (primary), and on the two co-primaries **ETA p90** and the **no-courier cancellation rate**.

Hypothesis. H_0 : treated = control zone-day fulfillment. H_1 : treated > control.

Method (both tracks). Unit = zone-day (420/arm, 12 zones). Three inference routes, all reported: - **A** — OLS `y ~ treat` with **cluster-robust (by zone) SE** (12 clusters — a floor, anti-conservative); - **B** — **randomization inference**: permute `arm` within (zone × weekday/weekend), 3,000 draws, two-sided p; - **C** — **wild-cluster bootstrap** (Rademacher weights on the null-model residuals, resampled by zone), 2,000 draws. Plus a covariate-adjusted OLS (`pre_fulfillment_rate` + `pre_liquidity_ratio` + `C(tier)`) for variance reduction, and the realised MDE for the design.

Evidence (`outputs/tables/03_primary.csv`, `03_power.csv`).

Metric	Control	Treatment	Effect	Cluster 95% CI	Cluster p	RI p	Wild-boot 95% CI (p)
Fulfillment rate	95.75%	97.61%	+1.87 pp	[+1.27, +2.46]	7×10^{-10}	0.0003	[+1.20, +2.55] (0.0015)
ETA p90 (min)	46.2	40.1	-6.06	[-8.56, -3.57]	2×10^{-6}	0.0003	[-9.25, -3.01] (0.002)
No-courier rate	1.16%	0.80%	-0.36 pp	[-0.64, -0.08]	0.011	0.031	[-0.67, -0.04] (0.023)

Covariate-adjusted fulfillment effect **+1.87 pp** ($p = 8 \times 10^{-10}$) — the pre-period covariates are zone-level and add no precision here. Design: 420/arm, zone-day SD 0.050, ICC ≈ 0.068 , design effect $\approx 5.7 \rightarrow$ **MDE $\approx +1.6$ pp** at 80% power; the observed +1.87 pp clears it.

Conclusion. The incentive **causes a real, statistically robust improvement in marketplace liquidity**: fulfillment **+1.87 pp**, ETA p90 **-6 min**, no-courier cancels **-0.36 pp**. The effect survives cluster-robust SEs, randomization inference and a wild-cluster bootstrap — the three routes agree, which matters with only 12 clusters. Whether it is *worth* the bonus cost is **04**.

Limitations. Fulfillment starts at $\sim 96\%$ in control, so the primary metric is ceiling-compressed — the effect reads more clearly on ETA p90 and no-courier rate, which is why they are treated as co-primary. 12 clusters is few; the RI p-value (assumption-light) is the one to anchor on.

04 — Guardrails: is it safe, and does it pay?

Question. G1 — does the bonus clear its cost per **incremental** delivered order? G2 — ETA. G3 — cancellation by cause. G4 — courier earnings.

Method (both tracks). G1: incremental delivered orders = (treated – control fulfillment) $\times$ treated orders placed, evaluated per zone against its matched control and **pooled** (Σ/Σ , so a zone with $\approx$ zero real lift contributes $\approx$ zero incremental rather than a clipped near-zero denominator); zone-cluster bootstrap of the pooled ratio (4,000 draws). Compared to the delivered-order contribution margin (**R\$ 10**, doc 01 \$9.4). G2–G4: OLS with cluster-robust SE.

Evidence (`outputs/tables/04_g1.csv` , `04_guardrails.csv`).

G1 — the economics.

Incentive spend (treated)	R\$ 35,820 on 7,446 delivered orders
Incremental delivered orders	≈ 127 — only 1.7% of the subsidised orders
Cost per incremental order	R\$ 281 (zone-bootstrap 95% CI [R\$ 195, R\$ 431])
vs contribution margin R\$ 10	fails by ~28× — CI nowhere near the margin
Value destroyed per subsidised order	≈ R\$ 4.6 (≈ the bonus itself)

G2–G4 (cluster-robust).

Guardrail	Control	Treatment	Effect	95% CI	p
ETA p50 (min)	36.8	33.9	–2.96	[–4.29, –1.63]	1×10^{-5}
ETA p90 (min)	46.2	40.1	–6.06	[–8.56, –3.57]	2×10^{-6}
Cancellation rate	4.25%	2.39%	–1.87 pp	[–2.46, –1.27]	7×10^{-10}
No-courier rate	1.16%	0.80%	–0.36 pp	[–0.64, –0.08]	0.011
Courier earnings / active hr	R\$ 33.0	R\$ 41.2	+R\$ 8.12	[+5.64, +10.59]	1×10^{-10}

Conclusion. Every **operational** guardrail is favourable — ETA down at both percentiles, all cancellation causes down, couriers earn materially more (the bonus plus more trips). But **G1 fails hard**: the bonus is a **flat payment on every delivery in the treated zone-block**, and ~98% of those orders would have been delivered anyway, so the R\$ 35,820 buys only ~127 extra orders at ~R\$ 281 each against a R\$ 10 margin. The lever improves the marketplace; its **pricing** is wrong.

Limitations. The contribution margin is a mid-band point value; a saved order also has retention value not counted here, but no plausible order value closes a 28× gap. The incremental-order count is small (127) so its bootstrap CI is wide.

05 — Novelty / decay: does the lift fade?

Question. Is the fulfillment lift a durable operating change, or a launch bump that decays as couriers stop reacting to the "surge zone" push?

Method (both tracks). From `fact_experiment_zone_block_week` (zone × experiment week × peak hour-block): the weekly treated-minus-control fulfillment lift; an OLS of that weekly lift on week number (slope test); the week-1 lift vs the settled (weeks 8–10) lift.

Evidence (`outputs/tables/05_weekly_lift.csv` , `05_decay.csv` , `outputs/figures/05_novelty.png`).

Week	1	2	3	4	5	6	7	8	9	10
Lift (pp)	+1.59	-0.17	-0.27	-0.17	-1.77	-0.72	+0.20	-1.47	+0.76	+0.48

Slope **-0.045 pp/week** ($p = 0.71$ — not significant). Week-1 lift **+1.59 pp**; weeks 8–10 mean ≈ 0 pp.

Conclusion. There is **no statistically detectable decay trend** — but the weekly series is very thin (each week is ~ 6 zones $\times$ ~ 4 peak blocks per arm), so the week-to-week lift swings by ± 1 – 2 pp on noise alone and the settled estimate is unreliable. Plan on the **pooled +1.87 pp from 03** (which uses all 840 zone-days); the weekly series is too noisy to either confirm a launch bump or rule one out. If the incentive were ever restructured and re-run, the design should carry more zone-days per week to resolve this.

Limitations. The block-week grain trades power for a time axis; the primary estimate does not depend on it.

06 — Heterogeneity: does it work everywhere, or only where supply is short?

Question. Is the fulfillment effect the same across the baseline supply-stress tiers (short / balanced / long), or concentrated where there is unmet demand to recover?

Method (both tracks). Per-tier fulfillment effect with cluster-robust (by zone) CIs. A formal **arm $\times$ tier interaction test**: OLS `fulfillment_rate ~ treat * C(tier)` (long = reference) with cluster-robust SEs — a Wald test on the two interaction terms — cross-checked with a nested-model F. A forest plot.

Evidence (`outputs/tables/06_per_tier.csv` , `06_interaction.csv` , `outputs/figures/06_forest.png`).

Tier	Zones	Control	Treatment	Effect	95% CI	p
short	6	94.44%	96.73%	+2.29 pp	[+1.54, +3.04]	2×10^{-9}
balanced	2	95.62%	98.05%	+2.43 pp	[+0.69, +4.18]	0.006
long	4	97.78%	98.72%	+0.94 pp	[+0.36, +1.53]	0.001

Arm $\times$ tier interaction: cluster-robust Wald $\chi^2 = 10.98$, $p = 0.0041$; nested-model F = 1.95, $p = 0.14$ (the two disagree — expected with 12 clusters and a 2-parameter Wald; the per-tier CIs are the cleaner evidence).

Conclusion. The effect is **real in all three tiers but roughly 2.5 $\times$ larger where supply is short or balanced** ($\sim +2.3$ – 2.4 pp) than where it is already long ($\sim +0.9$ pp). The long-tier CI [+0.36, +1.53] does not overlap the short-tier CI [+1.54, +3.04] — so "short > long" is clear even though the omnibus interaction test is borderline. Operationally: the incentive buys the most liquidity

exactly where the marketplace needs it, and near-nothing in the well-supplied zones — where it is **pure cost** (04 G1).

Limitations. Only 2 balanced-tier zones, so that estimate is wide. The tier labels are themselves derived from months 1–15 liquidity (`dim_zone`), not assigned.

07 — Neighbour-zone cannibalisation: is the local win partly a transfer?

Question. Does arming the bonus in one zone pull couriers out of adjacent (control) zones, so the metro-level gain is smaller than the sum of the local lifts? (A SUTVA violation — spatial spillover.)

Method (both tracks). - **A** — adjacent-control gap: fulfillment on control zone-days that *are* next to a treated zone that day vs control zone-days that are *not* (cluster-robust). - **B** — spatial regression on control zone-days: `fulfillment_rate ~ n_adjacent_zones_treated + pre_fulfillment_rate + C(zone) + C(dow)`, cluster-robust — coefficient = per-treated-neighbour drag. - **C** — clean-control bound: re-estimate the primary lift dropping control zone-days adjacent to any treated zone. - **D** — implied cannibalised share and the metro net. - **E** — supply elasticity: `available_courier_hours ~ treat`.

Evidence (`outputs/tables/07_cannibalisation.csv`).

	Estimate	95% CI / p
A. Adjacent-control fulfillment gap	−0.26 pp	p = 0.76 (not significant)
B. Per-treated-neighbour drag	−0.19 pp	p = 0.42 (not significant)
C. Primary lift: full-control vs clean-control	+1.87 pp → +1.64 pp	clean bound is <i>higher</i>
D. Implied cannibalised share	≈ 17%	metro net ≈ 83% of summed local lift
E. Supply response	+1.7 courier-hours / treated zone-day (p < 0.001); +0.6 delivered orders	

Conclusion. Cannibalisation is **present but small and not statistically distinguishable from zero** at this scale. Point estimates line up — control zones next to a treated zone fulfil ~0.2–0.3 pp lower, and the per-neighbour drag is negative — implying roughly **one-sixth** of the local gain is supply pulled from neighbours rather than net-new, so the metro-level effect is ~83% of the naive sum. The clean-control bound (+1.64 pp) is *close* to the full estimate, so depressed adjacent controls are not materially inflating the headline. The bonus does add real supply (+1.7 courier-hours/zone-day), it is just mostly paying couriers who were already there.

Limitations. 12 zones and a sparse adjacency graph give little power to resolve a ~0.2 pp spatial effect; the ~17% share is a point estimate with a wide implicit band. A denser design (more zones, an explicit "buffer" ring) would be needed to pin it down.

08 — Synthesis: the decision

Question. Should SwiftBite roll out the zone-hour incentive?

What the read-out establishes.

Dimension	Result	Route
Primary — fulfillment	+1.87 pp ([+1.27, +2.46])	cluster-robust + RI ($p = 3 \times 10^{-4}$) + wild-boot agree
Co-primary — ETA p90	−6.06 min ([−8.56, −3.57])	all $p < 0.002$
Co-primary — no-courier rate	−0.36 pp ([−0.64, −0.08])	$p \approx 0.01$ –0.03
Guardrail — cancellations (all causes)	−1.87 pp	favourable
Guardrail — courier earnings / active hr	+R\$ 8.12	favourable
Heterogeneity	short +2.29 · balanced +2.43 · long +0.94 pp	interaction Wald $p = 0.004$; short-CI $\cap$ long-CI = $\emptyset$
Decay	none detectable (series too thin)	slope $p = 0.71$
Cannibalisation	$\approx 17\%$ of local, n.s.; metro net $\approx 83\%$	adjacent-control gap $p = 0.76$
G1 — cost per incremental order	R\$ 281 ([R\$ 195, R\$ 431]) vs R\$ 10 margin	fails by $\sim 28\times$

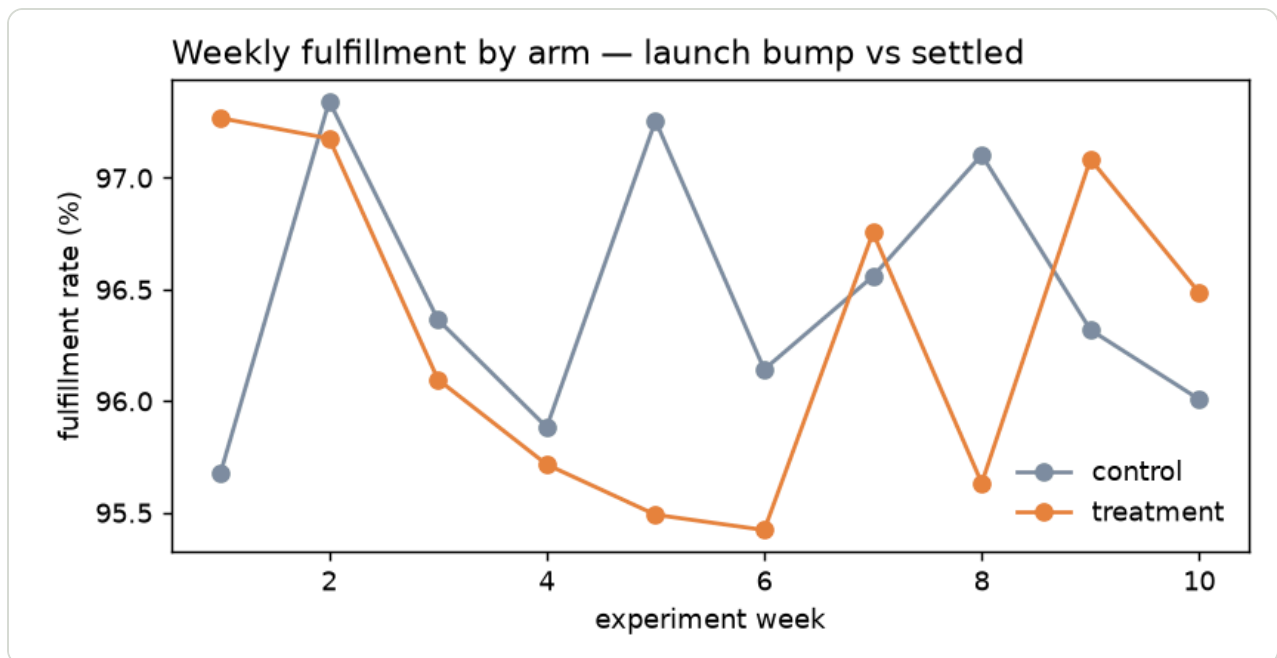
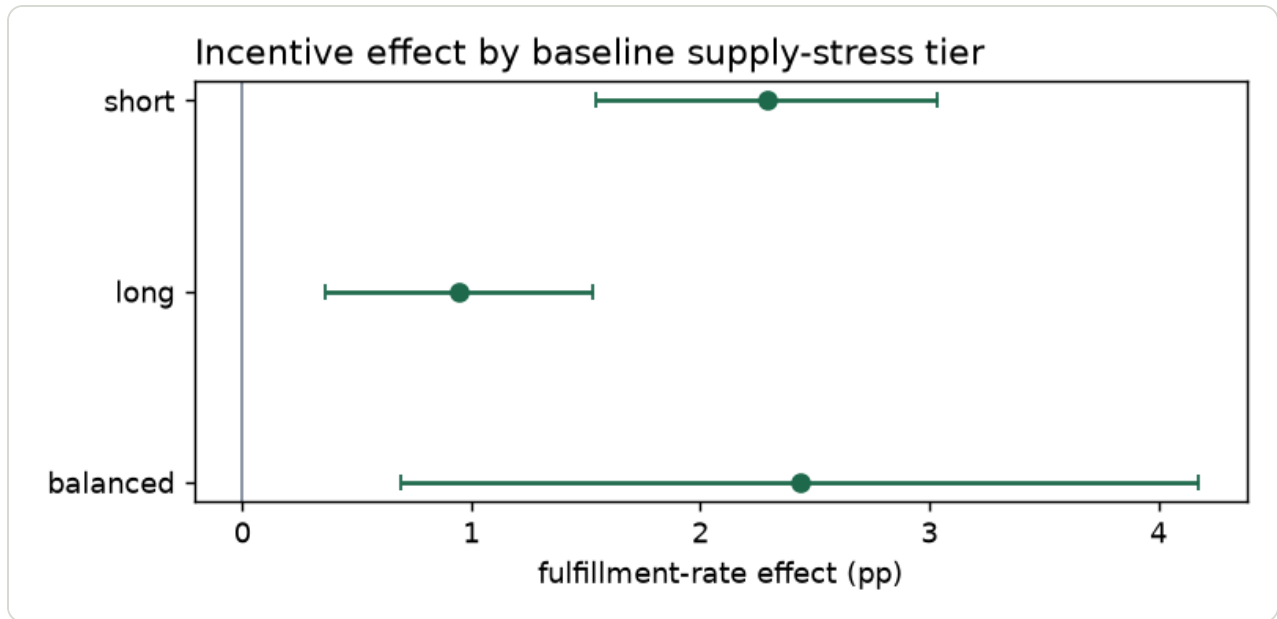
Recommendation.

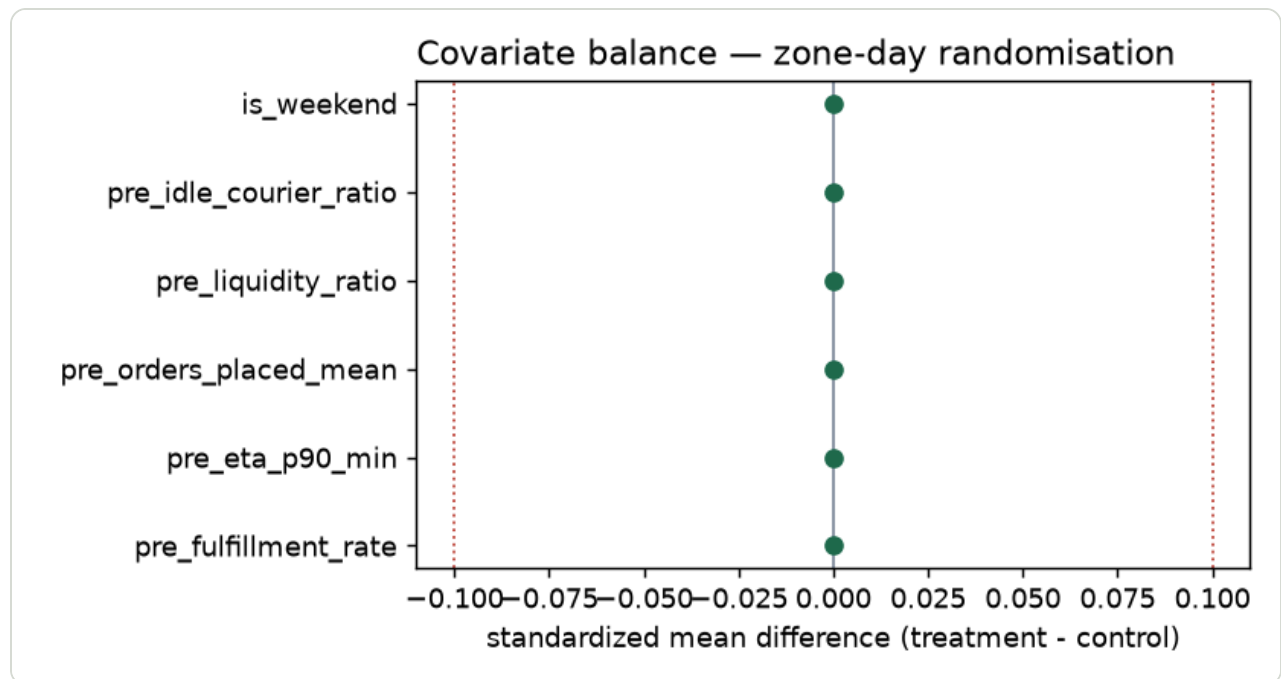
Do not roll out the flat per-delivery bonus as tested. It genuinely improves marketplace liquidity — fulfillment +1.9 pp, ETA p90 −6 min, no-courier −0.4 pp, concentrated in the supply-short and balanced zones, with only a small (~17%, not significant) neighbour-zone drag. **But** at ~R\$ 281 per *incremental* delivered order against a ~R\$ 10 contribution margin it destroys roughly R\$ 4.6 of value on every subsidised order: the bonus pays for all ~7,400 treated-block deliveries while only ~127 of them (1.7%) are incremental.

The lever works; its pricing is wrong. **Re-run only with the bonus restructured to reward incremental supply** — a threshold or surge-triggered payment that fires

only when a zone is actually short, not a flat per-delivery bonus — **capped to the supply-short and balanced zones**, with the neighbour-zone spillover guardrail and a hard budget live. Long (well-supplied) zones should never be in scope.

Why the experiment was still worth running. It cleanly separates two things ops conflated: the incentive *does* pull supply and clear demand (a real, robust, heterogeneous liquidity effect), and it *cannot* pay for itself in its current form (a pricing-design failure, not a lever failure). Both are actionable; neither was knowable from the dashboard.





Synthetic, seed-generated data (a portfolio project). Unit of randomisation = the zone-day (~840, 420/arm, 12 zones); every headline number is computed on two tracks (pandas + DuckDB) with a parity assertion — `python data_analysis/parity/run_all.py` — and reported with cluster-robust, randomization-inference and wild-cluster-bootstrap inference. The true effect lives only in `etl/config.py::INCENTIVE_EFFECT` and was recovered from the data. Per-analysis detail: `data_analysis/findings/01 - 08`.